

Quality of IE Preprocessing During preprocessing, we only keep high-confidence outputs from IE models to reduce errors. We observe this removes many of the noisy cases. To validate this, we manually evaluate the precision of each preprocessing step on a random sample of papers and observe that all steps yield high precision (91%-100%) except relation extraction (65%); in total, the rate of instances passing all steps was 79.7%.³

Gold Test Set We create a high-quality, clean test set. We remove test instances where models can trivially use surface-level background information to infer the ground truth to create a more *challenging set*, selecting instances with low similarity between background and ground truth sentences. We compute the cosine similarity between each instance’s background and corresponding ground truth sentence in the test set and take pairs with similarity ≤ 0.074 , which amounts to the tenth percentile of pairs. We further annotate this subset to create a *gold subset*. We manually exclude instances with trivial overlap between ground truth and background, remove cases with irrelevant background, and retain only instances where the target relation (from which the seed term is taken) is salient to the target sentence. We also remove test pairs that have unexplained terms in the background. We obtain a total of 194 instances.⁴

Split	Forward	Backward	Total
Train	55,884	58,426	114,310
Valid	7,938	8,257	16,195
Test	2,623	2,686	5,309

Table 1: Dataset statistics. Considering a relation of the form $[v \text{ used-for } u]$, we define $[v \text{ used-for } ?]$ as *forward*, and $[? \text{ used-for } u]$ as *backward*.

3 ScIMON Models

We present a new module to retrieve inspirations as contextual input (§3.1). Then, we describe another module to generate ideas given the context+inspiration (§3.2). Finally, we introduce a new iterative novelty optimization method to further improve idea quality (§3.3).⁵

3.1 Inspiration Retrieval Module

We take broad inspiration from cognitive aspects of innovation (Hope et al., 2023): when researchers generate a new idea, they are grounded in a web of existing concepts and papers bearing on the new idea. We aim to enrich the context of each background by retrieving “inspirations”—pieces of information that can guide hypothesis generation. As illustrated in Figure 2, for a given instance of the SCIMON task, our retrieval augmentation can retrieve from three types of sources. Each source uses a different form of query and output.

Semantic Neighbors For a given problem/motivation as input, ideas proposed for related problems in the training set can serve as a guiding reference for generating a new idea. Given the background context $\mathcal{B}$ with a seed term v and problem/motivation $\mathcal{M}$, we construct a base input b : a concatenation of $\mathcal{M}$ with a prompt $\mathcal{P}$ belonging to one of two templates: “ v is used for p ” or “ v is done by using p ”, where p is one of *Task/Method/Material/Metric*. In short, $b := \mathcal{P} \oplus \text{context}:\mathcal{M}$. For example, in Figure 2, the concatenation is “*Knowledge acquisition is done by using Method; Context:...requires plms to integrate information...lifelong manner...*”.

We then retrieve inputs from the training set that are semantically related to a new base input b , and obtain target sentences T corresponding to each retrieved training input. We extract the target term $t \in \mathcal{T}$ matching the seed term in b (§2.2) as inspiration for input b . Simply put, this means we use as inspiration the salient aspect of the solution proposed in $\mathcal{T}$, which we found empirically to help remove noisy/irrelevant information in $\mathcal{T}$. For example, in Figure 2, we find “*informative entities are done by using Method context: in this work, we aim at equipping pre-trained language models with structured knowledge.*” as similar to the input and use $t = \text{“linked knowledge graph”}$ as inspiration.

Technically, we first construct a fully connected graph $\mathcal{G}_S$ based on the training set where each node is a pair of input text b_i and target term t_i . We define the weight between two nodes i and j as the cosine similarity between b_i and b_j based on representations from SentenceBERT (Reimers and Gurevych, 2019) (all-mpnet-base-v2). Given b , we first insert it into $\mathcal{G}_S$ and compute the weights of its connected edges. We then retrieve neighbors input text $\{b_1, \dots, b_k\}$ from the training set with the largest edge weight, where k is the number of

³See Table 6 in Appendix.

⁴Full annotation details are in Appendix C.

⁵Training and hyperparameter details in Appendix B.

retrieved instances. We consider the corresponding target terms $\{t_1, \dots, t_k\}$ as semantic inspirations.

KG Neighbors We also explore enriching the context by linking it to a background KG with information on related methods and tasks. Using the same IE process used to extract our training examples (§2.2), we create a *global* background KG $\mathcal{G}_B$ which covers all papers in the corpus $\mathcal{D}_Y$ prior to a given year $\mathcal{Y}$ (i.e., the nodes in $\mathcal{G}_B$ correspond to tasks/methods/materials/metrics, and the edges are used-for relations, extracted and normalized from across the entire corpus as described earlier). Then, given a seed term v at query time, we select adjacent nodes $\{n_1, n_2, \dots\}$ from $\mathcal{G}_B$ as inspirations. As an example, in Figure 2, the neighbor nodes of “*knowledge acquisition*” include “*collaborative web text annotation editor*”, “*image matching*”, etc., which we select as inspirations.

Citation Neighbors Another notion of contextual relatedness we explore is via citation graph links. Here, given as input background context $\mathcal{B}$, we assume access to the original source document d from which $\mathcal{B}$ was extracted, and consider its *cited* paper title set $\mathcal{C}_d$ as potential candidates. This can be seen as a stronger assumption on information available to the model—assuming a researcher using the model provides relevant candidate documents from which ideas could be pooled. Because the training set only contains papers before year $\mathcal{Y}$, we only select papers $\mathcal{C}_{dY} \subseteq \mathcal{C}_d$ prior to year $\mathcal{Y}$. We then retrieve the top- k titles with the highest cosine similarity to d from $\mathcal{C}_{dY}$ based on their SentenceBERT embeddings as earlier. For instance, in Figure 2, the paper ELLE (Qin et al., 2022) cites the paper (de Masson d’Autume et al., 2019). Therefore, we choose the title “*episodic memory in lifelong language learning*” as inspiration information.

3.2 Generation Module

The idea generation module is given retrieved inspirations $i_1, \dots, i_k$ along with context $\mathcal{M}$ as input.

In-Context Learning We experiment with recent state-of-the-art LLMs, GPT3.5 davinci-003 (Ouyang et al., 2022) and GPT4 gpt-4-0314 checkpoint (OpenAI, 2023). We first ask the model to generate sentences based on the seed term and the context in the zero-shot setting without any in-context examples (GPT3.5ZS, GPT4ZS). We then ask the model to generate

sentences in a few-shot setting by prompting randomly chosen pairs of input and output from the training set (GPT3.5FS, GPT4FS). Inspired by Liu et al. (2022), we further employ a few-shot setting using semantically similar examples. Instead of random in-context examples, we use the top- k examples from the training set with the highest cosine similarity to the query (GPT3.5Retr). This few-shot retrieval setting differs from the semantic neighbor discussed above, in that we provide both the input and output of each instance rather than solely supplying target entities as additional input.

Fine Tuning We fine-tune T5 (Raffel et al., 2020) (more recent models may be used too; see our biomedical experiment §4.1 fine-tuning an LLM). We observe that the generation models tend to copy phrases from the background context. For example, given the context “...*hierarchical tables challenge numerical reasoning* ...”, the model will generate “*hierarchical table reasoning for question answering*” as the top prediction. For generating suggestions of *novel* ideas, we wish to discourage overly copying from the background context. We introduce a new in-context contrastive objective, where negative examples are taken from the text in the input (e.g., in Figure 2, the in-context negatives are *plms*, *pre-training*, etc). We compute an InfoNCE loss (Oord et al., 2018) over the hidden states of the decoder, aiming to maximize the probability of the ground truth against those of in-context negatives:

$$\begin{aligned} y^+ &= \sigma(\text{Avg}(\mathbf{W}_y \mathbf{h}^+ + \mathbf{b}_y)) \\ y_k^- &= \sigma(\text{Avg}(\mathbf{W}_y \mathbf{h}_k^- + \mathbf{b}_y)) \\ \mathcal{L}_{\text{cl}} &= \frac{\exp(y^+/\tau)}{\sum_k \exp(y_k^-/\tau) + \exp(y^+/\tau)} \end{aligned} \quad (1)$$

where $\mathbf{h}^+$ and $\mathbf{h}_k^-$ are decoder hidden states from the positive and k -th negative samples, $\mathbf{W}_y$ and $\mathbf{b}_y$ are learnable parameters, σ is a sigmoid function, τ is a temperature hyperparameter, and $\text{Avg}(\cdot)$ denotes the average pooling function based on the target sequence length. We optimize with both contrastive loss $\mathcal{L}_{\text{cl}}$ and the cross-entropy loss.

3.3 Iterative Novelty Boosting with Retrieval

We further improve the novelty of generated ideas with a new iterative *retrieve-compare-update* scheme. Conceptually, we consider a novelty-inducing penalty $\gamma_{\text{nov}}(\mathcal{I}, \mathcal{R})$ that penalizes ideas $\mathcal{I}$ that are too “close” to existing ideas in literature reference examples $\mathcal{R}$. $\gamma_{\text{nov}}(\mathcal{I}, \mathcal{R})$ is included

during in-context learning and inference, providing numerical feedback in the form of a score reflecting similarity to existing work. We wish to minimize this score while ensuring $\mathcal{I}$ remains relevant to the background context $\mathcal{B}$; we do so iteratively by (1) retrieving related work from $\mathcal{R}$, (2) measuring degree of novelty, (3) instructing the model to update $\mathcal{I}$ to be more novel w.r.t $\mathcal{R}$, conditioning on $\mathcal{B}$.

Specifically, in our implementation, we construct a reference corpus $\mathcal{R}$ based on all papers in the training set. We then propose an iterative algorithm that compares generated ideas against $\mathcal{R}$. We start with the initial idea $\mathcal{I}_0$ generated by the generation module. At each time step t , we use the generated idea $\mathcal{I}_t$ as a query to retrieve k nearest ideas from the literature reference corpus $\mathcal{R} = \{R_1, \dots, R_k\}$ based on SentenceBERT, with the top- k highest cosine similarity scores to $\mathcal{I}_t$ (we use $k = 20$). For each retrieved ground truth literature idea R_i , we compare its cosine similarity score S_i against a threshold μ (we use 0.6). We provide all the retrieved ground truth ideas $\hat{\mathcal{R}}$ that pass the threshold as additional negative examples for the large language models with the following instruction prompt: “*Your idea has similarities with existing research as demonstrated by these j sentences: $\mathcal{R}$. Make sure the idea you suggest is significantly different from the existing research mentioned in the above sentences. Let’s give it another try.*” We stop the iteration once all S_i are lower than μ . Figure 2 and Table 5 demonstrate novelty iterations.

4 Experiments

4.1 Human Evaluation

We present four human evaluation studies, exploring different facets of our problem and approach.

4.1.1 Study I: Comparing Outputs across Model Variants

We recruit six volunteer NLP experts with graduate-level education to rate the system. Raters are told to envision an AI assistant that suggests new paper ideas. We randomly select 50 instances (background+seed) from the gold subset. Each annotator receives ten instances, each paired with system outputs from different model variants (Table 2). We ask raters to assess idea quality by considering each output’s relevance to the context, novelty, clarity, and whether the idea is reasonable (positive ratings are dubbed “helpful” as shorthand, indicating they pass the multiple considerations). We observe

moderately high rater agreement.⁶ Raters are blind to the condition, and system outputs are randomly shuffled across instances.

We instruct annotators to only provide positive ratings to ideas sufficiently different from the input context. In Study I, we ask raters not to anticipate groundbreaking novelty from the system but rather a narrower expectation of quality and utility; in Study II below, we enrich the analysis to examine *ranking* between top models and also “raise the bar” and compare to actual ideas from papers.⁷

In a preliminary experiment, we also collected human ratings for GPT4-ZS (zero-shot) vs. GPT4-FS (few-shot) using the same criteria, finding GPT4-FS ranked higher in 65% of cases, with the rest mostly tied; thus, zero-shot GPT-4 was left out of the remainder of study I and subsequent studies to reduce annotation effort and cost.

Results Overall, GPT4FS and GPT4FS+KG outperform other models by a wide margin (Table 2). Apart from GPT4, T5+SN+CL performs best compared to other baselines, given its stronger prior knowledge of useful similar background hypotheses. In general, GPT3.5 models performed worse than fine-tuned T5 and its variants, which echoes results in other work in the scientific NLP domain (Jimenez Gutierrez et al., 2022). GPT4 outputs tended to be longer, which may partially explain higher human preference.

Type	3FS	3Rt	3FS+CT	3FS+KG	4	4+KG	T5	T5+SN
H	33	25	16	33	73	66	22	48
U	67	75	84	67	27	34	78	52

Table 2: Percent (%) of total votes each system output receives from human raters. H denotes a helpful output, while U denotes an unhelpful output. “3FS” refers to the GPT3.5FS. “3Rt” refers to the GPT3.5Retr. “4” refers to GPT4FS, and “4+KG” refers to the GPT4FS+KG. “T5+SN” refers to the T5+SN+CL. GPT4FS and GPT4FS+KG are rated much higher. While GPT4FS has a slightly higher rating than the KG variant, a further human study reveals that GPT4FS+KG often leads to more technical depth (§4.1).

4.1.2 Study II: Comparing GPT4 Variants against Real Papers

We conduct a follow-up human study of close competitors GPT4FS and GPT4FS+KG with a subset of

⁶The agreement scores are in Table 13 Appendix C.

⁷Full evaluator guidelines are in Appendix C. The sample annotations are in Table 11.